

Boundedness cannot be dropped in Theorem 3.5 of arXiv:2112.13900

Candidate counterexample packet

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Status: candidate full counterexample, likely valid. The literal open problem has a negative answer, already in the one-dimensional Hilbert space. The example is not quasi-bounded, so the stronger replacement question proposed in the next sentence of the source remains open.

1 Source problem

Adhikari–Aryal–Bhatt–Kunwar–Puri–Ranabhat prove the following annular existence theorem for a bounded, maximal monotone, positively homogeneous operator A ; its boundary assumptions are denoted (H3) and (H4) in the source [1].

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Theorem 3.5. *Assume that $G_1, G_2 \subset X$ are open, bounded with $0 \in G_2$ and $\overline{G_2} \subset G_1$. Let $L : X \supset D(L) \rightarrow X^*$ be linear maximal monotone with $\overline{D(L)} = X$, and $A : X \supset D(A) \rightarrow 2^{X^*}$ bounded, maximal monotone and positively homogeneous of degree $\gamma > 0$. Also, let $C : \overline{G_1} \rightarrow X^*$ be bounded, demicontinuous and of type (S_+) w.r.t. $D(L)$. Moreover, assume that*

- (H3) *there exists $v^* \in X^* \setminus \{0\}$ such that $Lx + Ax + Cx \not\supset \lambda v^*$ for all $(\lambda, x) \in \mathbf{R}_+ \times (D(L) \cap D(A) \cap \partial G_1)$, and*
- (H4) *$Lx + Ax + Cx + \lambda Jx \not\supset 0$ for all $(\lambda, x) \in \mathbf{R}_+ \times (D(L) \cap D(A) \cap \partial G_2)$.*

Then the inclusion $Lx + Ax + Cx \ni 0$ has a solution $x \in D(L) \cap D(A) \cap (G_1 \setminus G_2)$.

Proof. We begin by observing that a positively homogeneous and bounded maximal monotone operator A of degree $\gamma > 0$ satisfies $0 \in D(A)$ and $A(0) = \{0\}$. To solve the inclusion

$$Lx + Ax + Cx \ni 0, \quad x \in \overline{G_1}, \quad (3.10)$$

let us consider the associated equation

$$\hat{L}x + \hat{A}_t^\varphi x + \hat{C}x + tMx = 0, \quad t \in (0, \infty), \quad x \in j^{-1}(\overline{G_1}). \quad (3.11)$$

Here, the gauge function is $\varphi(r) = r^{p-1}$, $1 < p < \infty$, and $\gamma = p - 1$. We can show as in [5 Lemma 5] that there exists $R > 0$ such that the open ball $B_Y(0, R)$ contains all the solutions of (3.11). We recall that $Y = D(L)$.

Figure 1: Theorem 3.5 in the source PDF, page 17.

They then ask whether boundedness of A can be dropped, and suggest quasiboundedness as a possible replacement.

and the demicontinuity of C , we have $x_0 \in D(A)$, $w^* \in Ax_0$, and $Cx_n \rightarrow Cx_0$ in X^* . Thus, $Lx_0 + Ax_0 + Cx_0 \ni 0$.

Finally, in order to show $x_0 \in G_1 \setminus G_2$, we note that

$$G_R^1(Y) \setminus G_R^2(Y) = (G_1 \setminus G_2) \cap Y \cap B_Y(0, R) \subset G_1 \setminus G_2.$$

Consequently, $x_n \in G_1 \setminus G_2$ for all n , and therefore

$$x_0 \in \overline{G_1 \setminus G_2} \subset (G_1 \setminus G_2) \cup \partial(G_1 \setminus G_2) \subset (G_1 \setminus G_2) \cup \partial G_1 \cup \partial G_2.$$

By (H3) and (H4), $x_0 \notin \partial G_1 \cup \partial G_2$ and hence $x_0 \in D(L) \cap D(T) \cap (G_1 \setminus G_2)$. $\square$

3.1. Open Problem. Does Theorem 3.5 hold true if the boundedness of A is dropped? Since a positively homogeneous operator that is strongly quasibounded is necessarily bounded, it is desirable to determine whether Theorem 3.5 holds if A is assumed to be “quasibounded”. An operator $A : X \supset D(A) \rightarrow 2^{X^*}$ is said to be *quasibounded* if for every $S > 0$ there exists $K(S) > 0$ such that $\|x\| \leq S$ and $\langle x^*, x \rangle \leq S\|x\|$ for some $x^* \in Ax$ imply $\|x^*\| \leq K(S)$. The notions of quasibounded and strongly quasibounded operators were introduced in Hess [20].

4. APPLICATIONS

In this section, we apply Theorem 3.3 and Theorem 3.5 to elliptic and para-

Figure 2: Open Problem 3.1 in the source PDF, page 20.

2 Counterexample

Theorem 1. *The conclusion of Theorem 3.5 is false if the boundedness assumption on A is simply deleted. This holds for every prescribed degree $\gamma > 0$.*

Proof. Work in the real Hilbert space $X = \mathbb{R}$, identified with its dual. Fix $\gamma > 0$ and define the proper lower-semicontinuous convex function

$$\Phi(x) = \begin{cases} \frac{x^{\gamma+1}}{\gamma+1}, & x \geq 0, \\ +\infty, & x < 0. \end{cases}$$

Let $A = \partial\Phi$. Explicitly,

$$\text{dom } A = [0, \infty), \quad A(0) = (-\infty, 0], \quad A(x) = \{x^\gamma\} \quad (x > 0). \quad (1)$$

Because A is the subdifferential of a proper lower-semicontinuous convex function, it is maximal monotone. Formula (1) also verifies the source definition of positive homogeneity: for $s > 0$, $A(sx) = s^\gamma A(x)$, including $x = 0$ since every positive dilation preserves the negative ray; for $s = 0$, the required inclusion $0^\gamma A(x) \subset A(0)$ holds. The operator is not bounded in the source’s sense, because the bounded set $\{0\}$ has the unbounded image $A(0)$.

Now take

$$L = 0, \quad C = 0, \quad G_1 = (-2, 2), \quad G_2 = (-1, 1), \quad v^* = -1.$$

The zero operator L is linear maximal monotone with full domain. The map C is bounded and demicontinuous. It is of type (S_+) with respect to $\text{dom } L$ because weak and norm convergence coincide in $\mathbb{R}$. The sets are open and bounded, $0 \in G_2$, and $\overline{G_2} \subset G_1$.

It remains to check the two boundary hypotheses. Since

$$\text{dom } A \cap \partial G_1 = \{2\},$$

we have

$$L2 + A2 + C2 = \{2^\gamma\}, \quad \lambda v^* = -\lambda \leq 0 \quad (\lambda \geq 0).$$

Thus (H3) holds. Similarly, $\text{dom } A \cap \partial G_2 = \{1\}$ and the normalized duality map on $\mathbb{R}$ is $Jx = x$, so

$$L1 + A1 + C1 + \lambda J1 = \{1 + \lambda\} \not\equiv 0 \quad (\lambda \geq 0).$$

Hence (H4) holds.

Finally,

$$0 \in Lx + Ax + Cx = Ax \iff x = 0.$$

Indeed $0 \in A(0)$, while $A(x) = \{x^\gamma\} > 0$ for $x > 0$ and A is undefined for $x < 0$. But $0 \in G_2$, so $0 \notin G_1 \setminus G_2$. There is therefore no solution in $\text{dom } L \cap \text{dom } A \cap (G_1 \setminus G_2)$, contrary to the conclusion of Theorem 3.5. $\square$

3 Why this does not settle the quasibounded variant

The source defines A to be quasibounded when, for every $S > 0$, the conditions

$$\|x\| \leq S, \quad \langle x^*, x \rangle \leq S\|x\|, \quad x^* \in Ax$$

force a uniform bound on $\|x^*\|$. The operator (1) fails this at $x = 0$: every $x^* \leq 0$ belongs to $A(0)$ and satisfies the displayed pairing inequality, but $|x^*|$ is unbounded.

This pinpoints the first obstruction. Positive homogeneity makes $A(0)$ dilation-invariant. Quasiboundedness at zero would force $A(0)$ to be bounded, hence $A(0) = \{0\}$. The source proof also uses boundedness to obtain compactness of the Yosida approximants; those later uses are not repaired merely by assuming $A(0) = \{0\}$. Thus the packet proves exactly the negative answer to the literal first sentence, while making no claim about the stronger quasibounded replacement.

Five focused upgrade attempts are recorded in the accompanying attempt note. In particular, finite-dimensional quasibounded examples are structurally blocked, and the natural infinite-dimensional linear route collides with (H3). No later paper or explicit answer to this open problem was found in bounded exactPhrase, title, author, and structural searches through 12 August 2026.

4 Verification

The script `code/verify_counterexample.py` checks exact rational samples of monotonicity, the two boundary conditions, and absence of annular zeros. It reports

PASS: monotonicity samples, (H3), (H4), and absence of annular zeros.

The computational check is supplementary; maximal monotonicity follows analytically from the subdifferential representation.

References

- [1] D. R. Adhikari, A. Aryal, G. Bhatt, I. J. Kunwar, R. Puri, and M. Ranabhat, *Solvability of inclusions involving perturbations of positively homogeneous maximal monotone operators*, Electronic Journal of Differential Equations 2022, no. 63, 1–25; [arXiv:2112.13900](https://arxiv.org/abs/2112.13900).